

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 6 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|--------------------------------------|--|
| 1. Eigenvector B | (A) The scalar λ in $A\mathbf{x} = \lambda\mathbf{x}$: the factor by which its eigenvector is stretched. |
| 2. Eigenvalue A | (B) A nonzero vector $\mathbf{x}$ whose direction is unchanged by A — $A\mathbf{x}$ is just a multiple of $\mathbf{x}$. |
| 3. Characteristic equation C | (C) The equation $\det(A - \lambda I) = 0$, whose solutions are the eigenvalues. |
| 4. Diagonalization D | (D) Writing $A = X\Lambda X^{-1}$ with the eigenvectors in X and the eigenvalues down Λ . |
| 5. Diagonalizable matrix E | (E) A square matrix with enough (n) independent eigenvectors to fill the columns of X . |
| 6. Symmetric matrix F | (F) A matrix equal to its own transpose, $S = S^T$. |
| 7. Positive definite matrix G | (G) A symmetric matrix whose eigenvalues are all positive (its energy $\mathbf{x}^T S \mathbf{x} > 0$). |
| 8. Spectral theorem H | (H) The guarantee that a symmetric matrix has real eigenvalues and perpendicular eigenvectors. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- If $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, then $\mathbf{x}$ is an (A) **eigenvector** ← (B) pivot (C) determinant (D) inverse
- The eigenvalues of A are the solutions of (A) $A\mathbf{x} = \mathbf{0}$ (B) $\det A = 0$ (C) $\det(A - \lambda I) = 0$ ← (D) $A = A^T$
- The eigenvalues of a triangular matrix are (A) all 1 (B) **its diagonal entries** ← (C) its column sums (D) always 0
- A square matrix is diagonalizable exactly when it has (A) determinant 1 (B) a zero eigenvalue (C) equal rows (D) **n independent eigenvectors** ←
- A symmetric matrix $S = S^T$ always has (A) complex eigenvalues (B) **perpendicular eigenvectors** ← (C) determinant 0 (D) no eigenvectors
- The 2×2 symmetric matrix $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is positive definite when (A) $a < 0$ (B) $ac - b^2 = 0$ (C) $a > 0$ and $ac - b^2 > 0$ ← (D) $a = c$

Teacher Note

Item 1: an eigenvector is the direction A only stretches. Item 2: eigenvalues come from the characteristic equation $\det(A - \lambda I) = 0$. Item 3: a triangular (or diagonal) matrix wears its eigenvalues on the diagonal. Item 4: diagonalizable $\Leftrightarrow n$ independent eigenvectors (enough to fill X). Item 5: the spectral theorem — symmetric $\Rightarrow$ real eigenvalues *and* perpendicular eigenvectors. Item 6: the fast 2×2 positive-definite test, $a > 0$ and $ac - b^2 = \det > 0$.

Part C — Short Answer & Computation (35 pts)

Show your work.

- Eigenvector test for $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. (a) $A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ — yes, $\mathbf{x}$ is an eigenvector with eigenvalue $\lambda = 4$. (b) $A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, which is *not* a multiple of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (the direction changed), so $\mathbf{y}$ is not an eigenvector.
- Eigenvalues of $B = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$ by the characteristic equation. $\det(B - \lambda I) = \det\begin{bmatrix} 4-\lambda & 2 \\ 1 & 3-\lambda \end{bmatrix} = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda-2)(\lambda-5) = 0$, so $\lambda = 2$ and $\lambda = 5$.
- Eigenvectors of $B = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$ + trace/det check. $\lambda = 5$: $B - 5I = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}$, so $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. $\lambda = 2$: $B - 2I = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}$, so $\mathbf{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Check: trace $= 4 + 3 = 7 = 5 + 2$ (sum of λ 's); $\det B = 12 - 2 = 10 = 5 \cdot 2$ (product). ✓
- Compute A^2 for $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ using $A^2 = X\Lambda^2X^{-1}$. $X = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $\Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$, $X^{-1} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}$. Then $\Lambda^2 = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}$, and $A^2 = X\Lambda^2X^{-1} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ (matches squaring A directly). Only the λ 's were raised to the power.
- Diagonalizable or not? (a) $P = \begin{bmatrix} 3 & 0 \\ 1 & 3 \end{bmatrix}$ is triangular, so $\lambda = 3, 3$ (repeated). $P - 3I = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ gives only *one* eigenvector direction $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ — not enough to fill X , so P is not diagonalizable. (b) $Q = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has *distinct* eigenvalues $\lambda = 3, 1$, so it has two independent eigenvectors and is diagonalizable.
- Positive definiteness of $S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. (a) $a = 2 > 0$ and $ac - b^2 = (2)(2) - 1^2 = 3 > 0$, so S is positive definite (both eigenvalues $\lambda = 3, 1$ are positive). (b) S is symmetric ($S = S^T$), and by the spectral theorem a symmetric matrix always has perpendicular eigenvectors — no computation needed.
- System $\frac{d\mathbf{u}}{dt} = A\mathbf{u}$, $\lambda_1 = 3, \lambda_2 = -1$, $\mathbf{u}(0) = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$. (a) $c_1 + c_2 = 5$ and $c_1 - c_2 = 1 \Rightarrow c_1 = 3, c_2 = 2$. (b) $\mathbf{u}(t) = 3e^{3t}\begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2e^{-t}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. (c) Not stable: the mode with $\lambda_1 = 3 > 0$ has $e^{3t} \rightarrow \infty$, so $\mathbf{u}$ grows without bound (the growing mode wins); it does *not* go to 0.

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

- Why $\det(A - \lambda I) = 0$ finds the eigenvalues. (a) Rewrite $A\mathbf{x} = \lambda\mathbf{x}$ as $A\mathbf{x} - \lambda\mathbf{x} = \mathbf{0}$, i.e. $(A - \lambda I)\mathbf{x} = \mathbf{0}$. For λ to be an eigenvalue there must be a *nonzero* $\mathbf{x}$ that $A - \lambda I$ sends to $\mathbf{0}$. A matrix that crushes a nonzero vector to $\mathbf{0}$ has a nontrivial nullspace, so it is singular (not invertible). (b) A square matrix is singular exactly when its determinant is 0 (Unit 5). So the eigenvalues are precisely the numbers λ making $\det(A - \lambda I) = 0$ — solving that one equation hands us every eigenvalue.
- Powers by diagonalization; symmetric matrices. (a) $A^2 = (X\Lambda X^{-1})(X\Lambda X^{-1}) = X\Lambda(X^{-1}X)\Lambda X^{-1} = X\Lambda^2X^{-1}$, because the inner $X^{-1}X = I$ cancels; repeating gives $A^k = X\Lambda^kX^{-1}$. This is easy because Λ is diagonal, so Λ^k just raises each eigenvalue to the k — no repeated matrix multiplication. (b) By the spectral theorem, a symmetric matrix $S = S^T$ has real eigenvalues and eigenvectors that are automatically perpendicular. If in addition every eigenvalue is positive, then none is 0; since $\det S$ is the product of the eigenvalues, $\det S \neq 0$, so S is invertible.

Teacher Note

Scoring Part D (5 pts each). D1: 3 pts (a) $(A - \lambda I)\mathbf{x} = \mathbf{0}$ with $\mathbf{x} \neq \mathbf{0} \Rightarrow A - \lambda I$ singular; 2 pts (b) singular $\Leftrightarrow \det = 0$, so solve $\det(A - \lambda I) = 0$. D2: 3 pts (a) the $X^{-1}X$ cancellation gives $A^k = X\Lambda^kX^{-1}$, easy since Λ^k raises the λ 's; 2 pts (b) symmetric $\Rightarrow$ perpendicular eigenvectors (spectral theorem), all $\lambda > 0 \Rightarrow \det = \prod \lambda \neq 0 \Rightarrow$ invertible. Accept equivalent wording throughout.